

PORTULACARIA

branches. H and S 6–10ft (2–3m). Has ovate to rounded, fleshy, cream-edged, bright green leaves. In summer, bears small clusters of tiny, star-shaped, pale pink flowers. Z11–12 (min. 50°F/10°C)

POTAMOGETON

POTAMOGETONACEAE

Genus of deciduous, perennial, submerged water plants, grown for their foliage and as oxygenators. Suitable for cold-water pools and aquariums. Fully hardy. Prefers sun. Remove fading foliage and thin plants as necessary. Propagate by stem cuttings in spring or summer.

P. crispus (Curled pondweed) illus. p.442. **P. pectinatus** (Fennel-leaved pondweed). Deciduous, perennial, submerged water plant. S 10ft (3m). Has very narrow, linear, green to brownish-green leaves and, in summer, inconspicuous flowers. Suitable for a medium-sized to large pool. Z4–12

POTENTILLA

SYN. COMARUM

Cinquefoil

ROSACEAE

Genus of perennials, deciduous shrubs, and a few annuals and biennials, grown for their clusters of small, flattish to saucer-shaped flowers and their foliage. Tall species—particularly the shrubs—are useful in borders. Dwarf potentillas are good for rock gardens. Fully hardy. Does best in full sun, but flower color is better on orange-, red- and pink-flowered plants if shaded from the hottest sun. Needs well-drained soil. Propagate species by seed in fall; selected perennial forms by division only in spring or fall; shrubs by soft- or greenwood cuttings in summer.

P. alba illus. p.359.

P. arbuscula. See *P. fruticosa* Sulphurascens Group ‘Elizabeth’.

P. ‘Arc-en-ciel’ illus. p.268.

P. argyrophylla. See *P. atosanguinea* var. *argyrophylla*.

P. atosanguinea illus. p.269. **var. argyrophylla** (syn. *P. argyrophylla*) is a clump-forming perennial. H 1½ft (45cm), S 2ft (60cm). Saucer-shaped, yellow or yellow-orange flowers are produced in profusion from early to late summer above strawberrylike, silvery leaves. Z5–7

P. aurea illus. p.372.

P. crantzii (Alpine cinquefoil). Upright perennial with a thick, woody rootstock. H and S 4–8in (10–20cm). Has wedge-shaped, 5-lobed leaves and, in spring, bears flattish, yellow flowers with orange centers. Good in a rock garden. Z3–8

P. davurica var. *mandschurica* of gardens. See *P. fruticosa* ‘Manchu’.

P. eriocarpa illus. p.372.

P. ‘Etna’. Clump-forming perennial. H to 1½ft (45cm), S 2ft (60cm). Bears saucer-shaped, maroon flowers above strawberrylike, dark green leaves in midsummer. Z5–9

P. fragiformis. See *P. megalantha*.

P. fruticosa. Deciduous, bushy, dense shrub. H 3ft (1m), S 5ft (1.5). Dark green leaves have 5 to 7 narrowly oblong leaflets. From late spring to mid-fall, produces saucer-shaped, bright yellow flowers. Z3–7. **‘Abbotswood’** illus. p.149. **var. arbuscula** (syn. *P. arbuscula*), S 4ft (1.2m), bears golden-yellow flowers amid gray-

green to silver-gray leaves. **‘Beesii’** (syn. *P. ‘Nana Argentea’*), H 2ft (60cm), S 4ft (1.2m), is slow-growing and compact, with golden-yellow flowers and silver leaves.

‘Daydawn’ illus. p.162. **‘Elizabeth’** see *P.f. Sulphurascens* Group ‘Elizabeth’.

‘Friedrichsenii’ illus. p.160. **‘Gold Drop’** (syn. *P. parvifolia* ‘Gold Drop’), H and S 4ft (1.2m), bears abundant, golden-yellow flowers amid bright green leaves.

‘Goldfinger’ bears large, rich yellow flowers in profusion. **‘Jackman’s Variety’**, H 4ft (1.2m), has large, bright yellow flowers.

‘Maanelys’ (syn. *P. ‘Maanelys’*, *P. ‘Moonlight’*), H 4ft (1.2m), S 6ft (2m), has soft yellow flowers and gray-green foliage. **‘Manchu’** (syn. *P. davurica* var. *mandschurica* of gardens) has pure white flowers and silvery-gray leaves. **‘Red Ace’** illus. p.156.

‘Royal Flush’, H 1½ft (45cm), S 30in (75cm), produces mid-green leaves and sometimes semidouble, yellow-stamened, rich pink flowers, fading to white in full sun. **Sulphurascens Group ‘Elizabeth’** (syn. *P. arbuscula*, *P.f. ‘Elizabeth’*) illus. p.160. **‘Sunset’** illus. p.162. **‘Tangerine’**, H 4ft (1.2m), bears yellow flowers, flushed with pale orange-red, amid mid-green leaves. **‘Vilmoriniana’** illus. p.159.

P. ‘Gibson’s Scarlet’. Clump-forming perennial. H 1½ft (45cm), S 2ft (60cm). Bears saucer-shaped, brilliant scarlet flowers from mid- to late summer. Dark green leaves are strawberrylike. Z4–9

P. ‘Gloire de Nancy’, syn. *P. ‘Glory of Nancy’*. Clump-forming perennial. H 1½ft (45cm), S 2ft (60cm). Bears very large, saucer-shaped, semidouble, orange and coppery-red flowers throughout summer. Has strawberrylike, dark green leaves. Z4–9

P. ‘Glory of Nancy’. See *P. ‘Gloire de Nancy’*.

P. indica. See *Duchesnea indica*. **P. ‘Maanelys’**. See *P. fruticosa* ‘Maanelys’.

P. ‘Manelys’. See *P. fruticosa* ‘Maanelys’.

P. megalantha, syn. *P. fragiformis*, illus. p.276.

P. ‘Monsieur Rouillard’. Clump-forming perennial. H 1½ft (45cm), S 2ft (60cm). Produces saucer-shaped, double, blood-red flowers in summer above strawberrylike, dark green leaves. Z4–9

P. ‘Moonlight’. See *P. fruticosa* ‘Maanelys’.

P. ‘Nana Argentea’. See *P. fruticosa* ‘Beesii’.

P. nepalensis ‘Miss Willmott’ illus. p.265. **P. nitida**. Dense, mat-forming perennial. H 4in (10cm), S 6in (15cm). Has rounded, 3-lobed, silver leaves and, in summer, stems bearing 1–2 rose-pink flowers with dark centers. Often shy-flowering. Suits a rock garden or trough. Z4–9

P. parvifolia ‘Gold Drop’. See *P. fruticosa* ‘Gold Drop’.

P. recta. Clump-forming, hairy perennial. H 2ft (60cm), S 1½ft (45cm). Has toothed, gray- to mid- green, palmate leaves. From early to late summer, bears pale yellow flowers. Z4–8. **‘Macrantha’** see *P.r. ‘Warrenii’*. **‘Warrenii’** (syn. *P.r. ‘Macrantha’*) has rich, golden-yellow flowers throughout summer.

P. x tonguei. Mat-forming perennial. H 4in (10cm), S 1ft (30cm). Has rounded, 3- or 5-lobed, green leaves. Prostrate branches bear flattish, orange-yellow flowers with red centers during summer. Good for a rock garden. Z5–8

P. ‘William Rollisson’. Clump-forming perennial. H 1½ft (45cm), S 2ft (60cm). Has mid-green leaves. From early to late summer, bears saucer-shaped, semidouble, yellow or red-orange flowers with yellow centers and petal backs. Z4–8

P. ‘Yellow Queen’. Clump-forming perennial. H 1½ft (45cm), S 2ft (60cm).Has strawberrylike, dark green leaves. Produces bright yellow flowers from early to late summer. Z4–8

PRATIA

CAMPANULACEAE

Genus of prostrate, evergreen perennials grown for their abundant, solitary, star-shaped, 2-lipped, usually white or blue-purple flowers. Fully hardy to frost hardy. Good for groundcover in damp soil; also for a rock garden or in paving, but can be invasive. Grow in fertile, loamy, reliably moist soil in partial or deep shade. Propagate by division at any time.

P. angulata, syn. *Lobelia angulata*. Evergreen, creeping perennial. H 2in (5cm), S 1–2ft (30–60cm). Has small, broadly ovate, dark green leaves. Star-shaped, white flowers, with 5 unevenly spaced petals, are borne in leaf axils in late spring. Round, purplish-red fruits follow in fall. (borderline) Z7–10

P. pedunculata, syn. *Lobelia pedunculata*, illus. p.369.

PRIMULA

Primrose

PRIMULACEAE

Genus of mainly herbaceous perennials, some woody-based and evergreen. All have leaves in basal rosettes and tubular, bell- or primrose-shaped (flat) flowers. In some primulas, the flower stems, leaves, sepals, and, occasionally, sections of the petals are covered with a waxy powder known as farina. There are primulas suitable for almost every type of site: the border, scree garden, rock garden, peat garden, bog garden, pool margin, greenhouse, and alpine house. Some may be difficult to grow, as they dislike winter damp or summer heat. Fully hardy to frost tender. Repot container-grown plants annually. Tidy up fading foliage and deadhead as flowering ceases. Propagate species by seed when fresh or in spring; increase selected forms when dormant, either by division between fall and early spring, or by root cuttings in winter. Propagate auricula primulas by offsets in early spring or early fall. Border cultivars may be prone to slug damage in damp places and to attack by root aphids when grown in very dry conditions or in containers.

Primulas are divided into many different horticultural groups, of which the following are in common use. See also feature panel pp.258–259.

Auricula primulas

These are evergreen primulas, derived from hybrids between *P. auricula* and *P. hirsuta*, producing flat, smooth flowers borne in an umbel on a stem above the foliage. There are 3 main sub-groups: alpine, show, and border.

Alpine Auricula Group. In these, the color of the flower center is strikingly different from that of the petals. They may be either light-centered (white or pale in the center) or gold-centered (yellow or gold in the center). There is no meal or “farina” on either leaves or flowers. Grow in an alpine house or rock garden.

Show Auricula Group have flowers with a distinct circle of white meal or “paste” in the center. Some are self-colored, with one color, which may be red, yellow, blue or violet, from the central paste to the petal margins; edged cultivars have a black ring surrounding the central paste, feathering out to an often green, gray, or white margin; in fancy cultivars, the paste is surrounded by a color other than black, with a green, gray, or white margin. Show Auriculas have white farina on their foliage (except those with green-edged flowers), on their flower eyes and, sometimes, on their petal margins. Grow under glass to protect flowers from rain.

Border Auricula Group are generally robust, garden Auricula primulas, which are often very fragrant. Some have farina on flower stems and leaves. Grow in a mixed or herbaceous border.

Candelabra primulas

These are robust, herbaceous perennials with tubular, flat-faced flowers borne in tiered whorls up tall, sturdy stems. Some are deciduous, dying back to basal buds; others are semievergreen, dying back to reduced rosettes. Grow in moist shade or woodland, especially by streams.

Primrose-Polyanthus primulas

A diverse group of evergreen, semi-evergreen, or deciduous perennial hybrids, derived from *P. vulgaris*, crossed with *P. veris*, *P. juliae* and other species. They are divided into two main groups:

Primrose Group produce mostly solitary flowers among the leaves. Mainly grown as spring-flowering herbaceous perennials or as biennial, greenhouse-container plants flowering from winter to spring.

Polyanthus Group bear flowers in long-stalked umbels. Usually grown as biennials for bedding, sown in summer to flower in winter and the following spring, or under glass as winter- and spring-flowering container plants.

Cultivation

Primulas have varying cultivation requirements. For ease of reference, these have been grouped as follows:

- 1—Full sun or partial shade, in moist, but well-drained, humus-rich soil.
- 2—Partial shade, in deep, humus-rich, moist, neutral to acidic soil.
- 3—Deep or partial shade, in peaty, gritty, moist but sharply drained, acidic soil. Protect from excessive winter wet.
- 4—Under glass in an alpine house or frame. Avoid wetting foliage of mealy species and hybrids.
- 5—Full sun with some midday shade, or partial shade, in moist but sharply drained, gritty, humus-rich and slightly alkaline soil.
- 6—In a cool or temperate greenhouse, or as a houseplant, in bright, filtered light.